

Rachel Hoffman-Bice

CONTACT INFORMATION

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Massachusetts

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EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

Ph.D. Mechanical Engineering, GPA: 4.9/5.0 **June 2021**

Minor: Leadership and Management

Advisor: H. Harry Asada, Ford Professor of Engineering

Committee: H. Harry Asada, David Trumper, Alberto Rodriguez

S.M. Mechanical Engineering, GPA: 4.9/5.0 **June 2017**

Thesis: “A Multi-Track Elevator System for E-commerce Fulfillment Centers ”

Advisor: H. Harry Asada, Ford Professor of Engineering

University of Florida (UF), Gainesville, FL, USA

B.S. Mechanical Engineering, GPA: 3.97/4.0 **June 2015**

Minor: Electrical Engineering

Thesis: “A Method for Obtaining Formula One Race Track Curvature for use in Solving Minimum Lap Time Optimal Control Problems”

Advisor: Anil V. Rao, Erich Farber Faculty Fellow

PROFESSIONAL EXPERIENCE

Medtronic, Boston, MA, USA / Weßling, Germany
Robotic Surgical Technologies

February 2022 - Present

Senior Mechanical Design Engineer

June 2024 - Present

Senior Controls Software Engineer

January 2023 - June 2024

Controls Software Engineer II

February 2022 -January 2023

As a Mechanical Design Engineer

- Serve as Robotic Arm and Setup Arm (RASA) Hardware and Software Subject Matter Expert across the Surgical Robotics organization, leading design decisions, field failure investigations, testing and validation of both subsystems, with particular focus as hardware SME to multiple Hugo software release programs
- Technical lead for the next generation Control Arm (Insertion Axis (IA) and Instrument Drive Unit (IDU))— own requirements definition, technical drawings per ASME y14.5 and DFMEA-driven mechanical design, including tolerance stack analysis, FEA, loading conditions calculations, actuator performance. Lead design reviews validating hardware concepts against IEC 60601 and ANSI/AAMI HE75 requirements
- Own mechanical design of the next generation IA, including a cover to protect the control arm from water and contaminants while remaining cleanable, and selection of new cable connecting the IDU to the IA; iterate designs in Creo to meet cleanability and fluid ingress protection specifications
- Mechanical Technical Lead of a Robotic Arm actuator field brake failure Corrective and Preventative Action (CAPA), setting the investigation strategy and authoring comprehensive engineering reports including tolerance stack analysis, component change impact assessments with vendor testing, and failure cause map analysis; closed with fix implemented in field

- Supported next generation platform user studies in Boston and London as the hardware and software subject matter expert, including prototype preparation, on-site troubleshooting, shipment coordination and remote technical support, and summarized clinician feedback into design improvements
- Designed, built and tested a pin brake prototype for the next generation IA, including solenoid selection, pin and notched wheel design and brake engagement calculations, and led a workshop on the risks and benefits of pursuing a pin brake compared to a classic friction brake
- Built a departmentally-used materials selection tool(now in beta), that provides users with optimal material options along with information about preferred Hugo materials; developed the framework and training proposal for an in-house injection molding course
- Held technical ownership of the next generation IA primary actuator, IDU emergency release mechanism and hardware for localization and navigation, from concept generation through mechanical analysis, design review and prototype delivery to IEC 60601 requirements
- Extensively collaborated with suppliers and vendors to optimize component selection, evaluate design alternatives, and conduct testing and analysis to ensure product performance and manufacturability across the control arm subsystem
- Completed specialized training in GD&T Fundamentals, IEC-60601 medical device compliance, IEC-62304 software lifecycle, ISO-14971 risk management, PTC Creo Fundamentals and DRM (Six-Sigma) to enhance technical and leadership capabilities

As a Controls Software Engineer

- Serve as Robotic Arm and Setup Arm (RASA) software technical expert and Product Owner for the Surgical Robotics organization, authored and maintained documentation for the design, testing and validation required for the RASA software release, including 37 software items, 23 requirements and 18 design documents released into Agile, and led the team through maintenance and execution of 77 verification test case
- SAFe Certified Scrum team product owner of the RASA software (February 2022–July 2023), responsible for prioritizing system features and managing scrum team backlog focusing on RASA lower-level joint control and subsequent testing and validation; led team through design, verification and full release of one Hugo software release and participated in a second
- Authored software requirements and partnered with quality engineers to derive the DFMECA (per IEC 62304 and ISO 14971) and confirm software mitigations for identified risks were implemented, tested and captured in the design history file
- Re-architected the Instrument Drive Unit (IDU) controller for plug-and-play instrument support, incorporating software architecture team feedback and delivering a prototype that built and deployed to a complete robotic system within one Program Increment, then debugged it to full teleoperation with instrument exchanges before handing it off to the next development team
- Performed system investigations on the Wristed Ligasure robotic instrument to derive software and performance requirements, integrated a plant model into a simulator to enable Model-in-the-Loop testing without hardware, and performed system identification to develop a custom software controller for the instrument jaw
- Performed root cause analysis to determine the sources of RASA software failures on field systems, and developed and implemented risk control measures to mitigate potential RASA and instrument failure modes
- Wrote onboarding and knowledge transfer documentation on unit test development, signal logging and design document procedures, now adopted across the controls organization to support team growth and continuity
- Coordinated a staging-branch merge across seven scrum teams, owning the merge schedule and strategy, organizing code reviews, and hardware-testing sub-branches prior to integration

d'Arbeloff Laboratory For Information Systems and Technology, MIT, Cambridge, MA, USA

Visiting Engineer

June 2021- September 2022

- Provide expansive machine design expertise to lab members and consult on robotic prototype development and fabrication
- Designed gantry crane attachment for use with robotic crane system capable of manipulating a UR-5 Robot and 15kg aluminum shaft

Medtronic, Woburn, MA, USA

Research and Development Engineering Contingent

June 2021- February 2022

- Design, material selection and development of peristaltic pump to ensure intrauterine pressure does not exceed critical limits during minimally invasive hysteroscopic procedures such that regulatory compliance is met
- Performed extensive testing and selection of peristaltic pump tubing such that desired pressure and flow rate are maintained during operation
- Extensive usage of Solidworks to develop assembly drawings for export to packaging engineers for large scale manufacturing
- Conducted mechanical verification testing on minimally invasive surgical cutting tools to ensure resection performance met design specifications

MMID, Delft, Netherlands

Functionality Design Intern

June-August 2017

- Extensive concept development of a high-pressure pipe coupling, an apparatus for testing chemical dilution and a versatile electric outlet wall box, through use of verbal and non-verbal communication, including detailed hand-sketched concepts.
- Performed mechanical tests on wall mounted clamp prototypes to ensure material selection and design analysis met client specifications

MIT Lincoln Laboratory, Lincoln, MA, USA

Mechanical Engineering Intern, Systems and Analysis Group

June-August 2014

- Evaluated and tested strategies to develop optimal terrain following aircraft trajectory method. Designed and integrated terrain following MATLAB algorithm into radar testing software.
- Tested and analyzed results from improved radar testing software to ensure problem resolution

Cummins Inc., Columbus, IN, USA

Product Validation Intern

June-August 2013

- Analyzed the failure of a fuel filter and the implementation of preventative measures to ensure failure does not reoccur.
- Applied investigative problem solving methods to determine sources of failure for high pressure fuel lines, including resonant frequency and surface finish analysis

RESEARCH
EXPERIENCE

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

Ph.D. Research

2019 - 2021

“Precision Assembly of Underconstrained Heavy Shafts Suspended By Multiple Cables From A Robotic Crane”

d’Arbeloff Laboratory for Information Science and Systems, H. Harry Asada

- Investigated methods by which heavy shafts suspended with multiple cables at specific positions and orientations can be inserted into a chamfered hole despite a small ($120\mu m$) clearance
- Analyzed key conditions for successful peg insertion by analyzing different cable attachment

positions and orientations

- Designed and fabricated a scaled prototype of a three-winch system with linear actuators designed to manipulate a 15 kg payload, including software and control of the system
- Analyzed system force balance to determine the required balance of cable tension to prevent the peg from jamming during the insertion process and to correct for wedging using cables
- Software and Tools used: Solidworks, MATLAB, Python, Pyserial, Odrive, OpenCV, April-Tag, ArucoTag, Harmonic Drive brushless motors, Absolute Rotary Encoders, Stepper Motors, Arduino, Load Cells, HX711 Load Cell Amplifiers, Stratasys Dimension 1200es 3D printer, Mark-forged 3D printer

Master's Research

2015 - 2017

"A Multi-Track Elevator System for E-Commerce Fulfillment Centers"

d'Arbeloff Laboratory for Information Science and Systems, H. Harry Asada

- Invented a novel mechanism by which elevators can travel along tracks, and transfer between track segments of varying direction. This mechanism was inspired by a railway turnabout and the rack-pinion drive system of cog-railways.
- Analyzed optimal scheduling routes for multiple carts using this system and detailed the advantages over current elevator systems in e-commerce fulfillment centers.
- Designed and manufactured/implemented scale model consisting of thirteen direction changing mechanisms and twenty-seven track segments
- Software and Tools used: Solidworks, MATLAB, Dynamixel Smart Servo Motors, Waterjet, Table Router

University of Florida (UF), Gainesville, FL, USA

Undergraduate Thesis

2014 - 2015

"A Method for Obtaining Formula One Race Track Curvature for use in Solving Minimum Lap Time Optimal Control Problems"

Vehicle Dynamics and Optimization Laboratory, Anil V. Rao

- Developed user-interactive MATLAB image processing software to obtain formula one track curvature and parametric equations for use in optimal control problems solved in GPOPS-II software
- Allows for users to upload an image of a desired formula one track for analysing minimal lap time

TEACHING
EXPERIENCE

Massachusetts Institute of Technology, Cambridge, MA, USA

Laboratory Teaching Assistant

Spring 2020

2.12/2.120: Introduction to Robotics, H. H. Asada, K. Youcef-Toumi, H. Chin

- Led TAs and Undergrad LAs in providing instruction during laboratory teaching sessions.
- Consulted and mentored two student-teams on the design of a three degree of freedom robotic arm specific to the final project challenge of sorting and throwing plastic bottles to be recycled.
- Provided mentorship to the students through teaching them robot mechanism design and machine design fundamentals and good practices
- Aided the instructors in administration, grading, and laboratory instruction
- Implemented Robotic MATLAB simulations, involving kinematics, trajectory planning and object manipulation, for students continuing education at home due to COVID-19

HONORS AND
AWARDS

Third Place, MIT Mechanical Engineering de Florez Award Competition

2021

Mechanical Engineering Rising Stars Workshop, Berkeley CA

2020

Winner, Best Application Paper Award, IEEE IROS

2017

“A Multi-Track Elevator System for E-Commerce Fulfillment Centers”

Honorable Mention, National Science Foundation Scholarship **2016**

Graduated *summa cum laude*, University of Florida **2015**

Winner, Outstanding Scholar Commencement Award, University of Florida **2015**

Winner, Fyfe Family Foundation Scholarship, University of Florida **2014**

Winner, Pratt and Whitney Scholarship, University of Florida **2013**

MENTORSHIP AND
SERVICE

Baker House, Cambridge, MA, USA

Graduate Resident Advisor

2016 - 2021

- Foster an inclusive, respectful, supportive, safe, and positive living environment through mentoring, educating, encouraging personal growth, providing outlets for managing stress, and facilitating positive interpersonal relationships.
- Coordinated the Resident Peer Mentor within an undergraduate residence hall and organized and structured inaugural group of students selected to be departmental peer advisors.

MIT Medical Device Design Course, Cambridge, MA, USA

Mentor

2017 - 2019

Provide extensive mechanical design consulting for student-lead medical device design teams, including the development of a Crank-Powered Otoscope, a Center-of-Gravity Detector for Heavy Machinery Operator Fatigue, a Lower Extremity Fracture Fixation Device, a Preterm Vaginal Breech Delivery Device and a Portable Device for helping the physically impaired stand.

UF Peer Advisors Program, Gainesville, FL, USA

Student Organizer, Mentor

2014-2015

Coordinated with faculty to introduce the Peer Advising and Peer Mentor programs into the Mechanical and Aerospace Engineering Department. I was responsible for guiding students through the course selection process and advising them about clubs, available resources, research opportunities and suggested course loads.

MIT Women’s Initiative, Cambridge, MA, USA

Participant

January 2018

Traveled to Detroit, MI to speak at 10 different public elementary and middle schools about the value of women in STEM

Pi Tau Sigma, University of Florida Chapter, Gainesville, FL, USA

Member

2010-2012

Coordinated Undergraduate Research Symposium

OTHER PROJECTS
AND CAMPUS
INVOLVEMENT

Design of a Disinfection Robot, Cambridge, MA, USA

d’Arbeloff Laboratory for Information Systems and Technology

2020

Design and Fabrication of a mobile platform capable of driving around areas on campus and using high-powered UV disinfection lights to cleans areas periodically to prevent the spread of Covid-19

Robotic Assisted Wire Harness Installation Demonstration Effort , Cambridge, MA, USA

d’Arbeloff Laboratory for Information Systems and Technology

2018-2019

- Developed a control strategy coordinating a dual-arm robotic system for positioning wire harness while avoiding collisions
- Implemented Dual-Arm Impedance and Admittance Control for safe human-robot interaction
- Developed a custom path planning software using Gazebo, two sawyer robots and pybullet physics simulator for collision detection
- Software and Tools used: Solidworks, Python, ROS, Robot Raconteur, Pyserial, Gazebo, Pybullet, Robotiq 2F-85 Grippers

Design of a T-Based Lathe, Cambridge, MA, USA

MIT Precision Machine Design Course

2018

Deterministic design, fabrication and testing of a custom T-Based Lathe. The parts cur on the lathe measured flatness 187.9 um +/- 5.75 um and the Error Budget Predicted flatness 233 um +/- 22.7um

Control System for Recreating realistic human reaching movements in a direct drive robot arm, Cambridge, MA, USA

Biomechanics and Neural Control of Movement Course

2016

Development of a 2DOF direct drive robot arm with control system representing joint-level reflex feedback, musculoskeletal simulation and the translation of those signals into torque values commanded to the robot.

Design of a Foldable Electric Scooter, Gainesville, FL, USA

Capstone Project

2018

Design and fabrication of a foldable electric scooter capable of carrying a 220 lb adult. The scooter can reach 15.5 mph top speed with a 1.5 s stop time.

DrumBot, Gainesville, FL, USA

Intelligent Robots and Machines Lab

2015

Design, fabrication and programming of a mobile platform with a miniature drum kit attached. The motions of the user are captured by an XBOX Kinect and translated into motions of drumsticks attached to the drumkit.

ADDITIONAL SKILLS Controls and Software: MATLAB, Simulink, Python, C/C++ ,controller design and debugging; system identification; Model-in-the-Loop test development; unit testing; ROS, OpenCV, Arduino, Odrive

Mechanical and Hardware: PTC Creo (Multibody, 2D Drawing, Analysis, Kinematic Connections), SolidWorks, Ansys; GD&T and tolerance stack analysis; DFM/DFA; DFMEA; actuator, brake and mechanism design; prototyping and bench test fixture development

Languages: Python, MATLAB

Applications: Microsoft Office Suite/Teams/SharePoint, Gitlab, Github, Jira, Confluence, Miro, Slack, VSDC Video Editor

Manufacturing and Tools : Mill, Lathe, Waterjet, Laser Cutter, 3D printers, Hand Tools, HAAS CNC Mill

CITIZENSHIP United States of America

REFERENCES Available upon request.

JOURNAL PUBLICATIONS **R. Hoffman** and H. H. Asada, "Control Strategy for Jam and Wedge-Free 3D Precision Insertion of Heavy Objects Suspended with a Multi-Cable Crane," *IEEE Robotics and Automation Letters (RA-*

L), vol. 6, no. 4, pp. 7453-7460, Oct. 2021, doi: 10.1109/LRA.2021.3093860. *The contents of this paper were also selected by the IEEE International Conference on Intelligent Robots and Systems (IROS) 2021 Program Committee for presentation at the Conference in Prague, Czech Republic (virtual).*

R. Hoffman and H. H. Asada, "Precision Assembly of Heavy Objects Suspended with Multiple Cables From a Crane," *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, no. 4, pp. 6876-6883, Oct. 2020, doi: 10.1109/LRA.2020.3013845. *The contents of this paper were also selected by the IEEE International Conference on Intelligent Robots and Systems (IROS) 2020 Program Committee for presentation at the Conference in Las Vegas, Nevada, USA (virtual).*

J. Guggenheim, **R. Hoffman**, H. Song and H. H. Asada, "Leveraging the Human Operator in the Design and Control of Supernumerary Robotic Limbs," *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, no. 2, pp. 2177-2184, April 2020. *The contents of this paper were also selected by the IEEE International Conference of Robotics and Automation (ICRA) 2020 Program Committee for presentation at the Conference in Paris, France. (virtual)*

Whalen, M., Chang-Davidson, E., Moran, T., **Hoffman, R.**, Frydman, G., Slocum, A., and Dangel, A., "Device Prototype for Vaginal Delivery of Extremely Preterm Fetuses in The Breech Presentation," *ASME. J. Med. Devices*, November 11, 2020, doi: <https://doi.org/10.1115/1.4049086>.

CONFERENCE
PUBLICATIONS

R. Hoffman, and H.H. Asada, "A Multi-Track Elevator System for E-Commerce Fulfillment Centers," *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2017)*, Vancouver, Canada, October 2017.

J. Chacon-Castano, D. Rathone, **R. Hoffman**, H. Yang, D. Pantazis, J. Yang and N. Hanumara, "Music and the Brain – Design of an MEG Compatible Piano," *2017 IEEE International International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2017)*, Seogwipo, South Korea, July 2017.